Java Lab Program - 1

1a)Write a Java program to find the roots of quadratic equation $ax^2+bx+c = 0$

Note : your program must satisfy the following test cases :

Type	Input	Expected Output	Explanation
Success	a=1, b=-5, c=6	Roots = 3, 2	Real roots
Success	a=1, b=2, c=1	Roots = -1, -1	Equal roots
Success	a=1, b=0, c=-4	Roots = 2, -2	Positive and negative roots
Failure/Edge	a=1, b=1, c=1	Imaginary roots	Negative discriminant
Failure/Edge	a=0, b=2, c=3	Not quadratic	a cannot be zero

AIM

To solve the quadratic equation $ax^2 + bx + c = 0$, the algorithm should first check whether the equation is valid ($a \neq 0$), then calculate the discriminant and determine the type of roots.

Algorithm: Find the Roots of a Quadratic Equation

1. Start.
2. Input the coefficients a, b, and c.
3. If a = 0:
    - Display "Not quadratic".
    - Stop.
4. Compute the discriminant:
    - D = b × b − 4 × a × c
5. If D > 0:
    - Calculate:
        - root1 = (-b + √D) / (2 × a)
        - root2 = (-b − √D) / (2 × a)
    - Display both roots.
6. Else if D = 0:
    - Calculate:
        - root = -b / (2 × a)
    - Display the equal roots.
7. Else (D < 0):
    - Display "Imaginary roots".
8. Stop.

Program

```
java.util.Scanner;

public class QuadraticRoots {
   public static void main(String[] args) {
      Scanner sc = new Scanner(System.in);

      System import.out.print("Enter a: ");
      double a = sc.nextDouble();

      System.out.print("Enter b: ");
      double b = sc.nextDouble();

      System.out.print("Enter c: ");
      double c = sc.nextDouble();

      if (a == 0) {
         System.out.println("Not quadratic");
      } else {
         double discriminant = b * b - 4 * a * c;

         if (discriminant < 0) {
            System.out.println("Imaginary roots");
         } else {
            double root1 = (-b + Math.sqrt(discriminant)) / (2 * a);
            double root2 = (-b - Math.sqrt(discriminant)) / (2 * a);

            // Display integer values without decimal points when
possible
            if (root1 == (int) root1 && root2 == (int) root2) {
               System.out.println("Roots = " + (int) root1 + ", " + (int)
root2);
            } else {
               System.out.println("Roots = " + root1 + ", " + root2);
            }
         }
```

```
        }

        sc.close();
    }
}
```

Output:

```
D:\JAVALAB>javac QuadraticRoots.java

D:\JAVALAB>java QuadraticRoots
Enter a: 1
Enter b: -5
Enter c: 6
Roots = 3, 2

D:\JAVALAB>java QuadraticRoots
Enter a: 1
Enter b: 2
Enter c: 1
Roots = -1, -1

D:\JAVALAB>java QuadraticRoots
Enter a: 1
Enter b: 0
Enter c: -4
Roots = 2, -2

D:\JAVALAB>java QuadraticRoots
Enter a: 1
Enter b: 1
Enter c: 1
Imaginary roots

D:\JAVALAB>java QuadraticRoots
Enter a: 0
Enter b: 2
Enter c: 3
Not quadratic

D:\JAVALAB>
```

1b)Write a Java program to print prime numbers up to a given integer (use BufferedReader)

Note : your program must satisfy the following test cases :

Type	Input	Expected Output	Explanation
Success	N=10	2 3 5 7	Normal case
Success	N=20	2 3 5 7 11 13 17 19	Multiple primes
Success	N=2	2	Smallest prime
Failure/Edge	N=1	No prime numbers	Boundary case
Failure/Edge	N=-5	Invalid input	Negative value

AIM

print prime numbers up to a given integer (use BufferedReader)

Algorithm: Print Prime Numbers up to a Given Integer (Using BufferedReader)

1. Start.
2. Read the integer N using BufferedReader.
3. If N < 0:
    - Display **"Invalid input"**.
    - Stop.
4. If N < 2:
    - Display **"No prime numbers"**.
    - Stop.
5. For each number i from 2 to N:
    - Assume i is prime.
    - For each integer j from 2 to √i:
        - If i % j == 0:
            - Mark i as **not prime**.
            - Exit the inner loop.
    - If i is prime:
        - Print i.
6. Stop.

Program

```
import java.io.BufferedReader;
import java.io.IOException;
import java.io.InputStreamReader;

public class PrimeNumbers {
    public static void main(String[] args) throws IOException {
        BufferedReader br = new BufferedReader(new
InputStreamReader(System.in));

        System.out.print("Enter N: ");
        int N = Integer.parseInt(br.readLine());

        if (N < 0) {
            System.out.println("Invalid input");
        } else if (N < 2) {
            System.out.println("No prime numbers");
        } else {
            for (int i = 2; i <= N; i++) {
                boolean isPrime = true;

                for (int j = 2; j <= Math.sqrt(i); j++) {
                    if (i % j == 0) {
                        isPrime = false;
                        break;
                    }
                }

                if (isPrime) {
                    System.out.print(i + " ");
                }
            }
        }
    }
}
```

Output:

```
D:\JAVALAB>javac PrimeNumbers.java

D:\JAVALAB>java PrimeNumbers
Enter N: 10
2 3 5 7
D:\JAVALAB>java PrimeNumbers
Enter N: 20
2 3 5 7 11 13 17 19
D:\JAVALAB>java PrimeNumbers
Enter N: 2
2
D:\JAVALAB>java PrimeNumbers
Enter N: -5
Invalid input

D:\JAVALAB>
```